

26 West Slew Speed Project Report

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Abstract

When lightning struck 26 West, one of PARI's 26-meter radio telescopes built for NASA's Rosman Tracking Station and later used by the Department of Defense during the Cold War, the entire servo drive system had to be replaced. The new system was commissioned with a conservative slew-speed limit of $0.33^\circ/\text{sec}$, far below the telescope's original hydraulic-era capability of $3^\circ/\text{sec}$, with no formal analysis to establish where the true safe limit lay.

This project aims to determine the maximum safe slew speed for both axes of 26 West under the new servo drive system. The central constraint is stopping distance: the telescope must be able to come to a complete stop within 2° of travel, representing the physical buffer between the software pre-limit switches and the mechanical dead-limit switches on each axis.



Figure 1: 26 West

Setup

The basic idea hinges on 2 basic kinematic equations. This is the first of the two:

$$\tau = I\alpha$$

According to the equation, the torque provided to move an object is equal to the moment of inertia of the rotating body multiplied by its angular acceleration. We can use this concept to find the maximum angular acceleration of the dish about each axis by using the highest safe torque that can drive the telescope and dividing that number by a conservative estimate of the dish's inertia. The inertia estimate can be found by making basic geometric inertia models based on blueprints of the dish and conservative mass estimates. Once the angular acceleration is found, it can be plugged into the second equation:

$$\omega_f^2 = \omega_i^2 + 2\alpha\theta$$

Which, if we assume our final angular velocity to be 0, can be rearranged into this:

$$\omega_i = \sqrt{(2\alpha\theta)}$$

According to this equation, the initial angular velocity of an object is equal to the square root of twice its angular acceleration multiplied by its change in angle. Applied, it means that if our newfound angular acceleration is multiplied by our change in angle (2°), doubled, then the square root is taken, out pops the maximum safe slew speed for the telescope.

This process is repeated for both axes and for three different torque assumptions, producing a range of results from most to least conservative:

$\frac{1}{2}\omega_s$ - rated torque of a single servo motor. Two motors drive each axis, but they are sometimes configured to work against each other to combat gear lash, effectively halving the available net torque.

ω_s - rated torque of both servo motors contributing fully in the same direction.

ω_m - mechanical brake torque, listed in the operating manual as five times rated motor torque. This is the do-not-exceed ceiling, representing the speed from which the emergency brakes can still stop the dish within 2° .

Inertia Models

Completing the calculation requires a moment of inertia estimate for the telescope about each axis. Because the telescope was built in the early 1960s, detailed structural documentation is scarce. What is available includes the original 1964 blueprints, providing scale drawings and dimensions for many components, and the pier load documentation, which gives the maximum dead weight the ground supports

were designed to carry. That dead load rating serves as a conservative upper bound on total telescope mass. A heavier assumed mass produces a higher inertia, a lower calculated safe speed, and a safer result. This principle of deliberate conservatism is applied throughout all mass and inertia estimates in this project.

The telescope moves about two perpendicular axes and can be divided into three systems: the Y-axis system (the dish assembly and Y-axis counterweights), the X-axis system (the X-axis counterweights), and the fixed support structure. This requires three separate inertia calculations: the Y-system about the Y-axis, the Y-system about the X-axis, and the X-system about the X-axis.

The mass relationships between components were established using a 2006 capstone project report from North Carolina Agricultural and Technological State University, in which students physically balanced both axes of 26 West by calculating the moment imbalance and adjusting the counterweights to approach equilibrium. The report notes that true neutral balance is not desirable for a telescope of this type. A slightly counterweight-heavy configuration is preferred so that the dish naturally returns to zenith when unpowered, resisting wind loads. From the report, the amount of mass required to achieve neutral balance and the amount actually removed allowed the residual moment imbalance to be calculated, establishing the moment relationship between the dish and its counterweights. Combined with counterweight positions scaled from the drawings, the mass relationships between components could be determined.

With those relationships established, the dish assembly was modeled in SolidWorks using the original blueprints as reference, though simplified considerably. The structural array supporting the dish from behind consists of angle iron members; for modeling efficiency, these were represented as solid circular beams with densities adjusted according to the ratio of the angle iron cross-sectional area to the circular cross-sectional area, preserving the correct mass in the correct location. This density-adjustment strategy was applied wherever geometry was simplified, because at this scale the specific shape of a component matters far less to the inertia calculation than its mass and its distance from the axis. This is captured by the parallel axis theorem:

$$I = I_{cm} + md^2$$

The total inertia of an object rotating about an axis outside its center of mass equals its self-inertia plus its mass multiplied by the square of its distance from the axis. For the large, widely distributed masses in this telescope, the md^2 term dominates by orders of magnitude, making precise shape modeling far less critical than accurate mass placement.

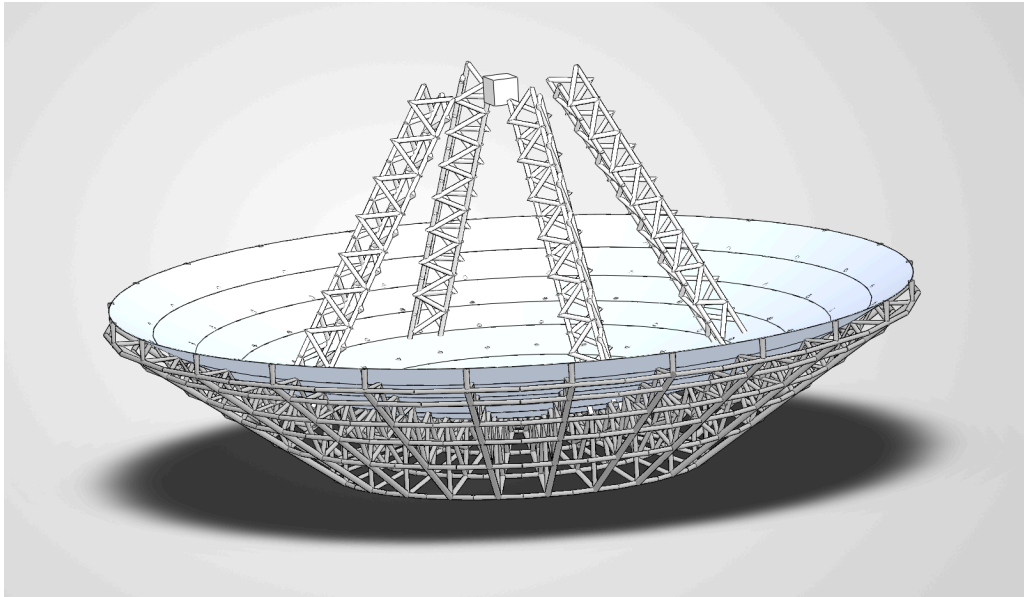


Figure 2: 26W SolidWorks Model

Once the dish assembly mass was established in SolidWorks, the counterweight masses were derived from the moment relationships. All component masses were then raised by 20% before being used in further calculations. The Y-axis counterweights were modeled as two rectangular prisms placed at the known counterweight radius; the X-axis counterweights were modeled as a single triangular prism. After a further 20% increase applied to the final inertia values, the total moments of inertia were:

X-axis (major axis): 29,046,559 kg·m²

Y-axis (minor axis): 7,812,955 kg·m²

Results

Substituting these inertia values, the gear ratios, and the three torque cases into the equations above produced the following maximum safe slew speeds:

X/Major-axis:

$$\frac{1}{2} \omega_s = 1.92 \text{ deg/sec}$$

$$\omega_s = 2.71 \text{ deg/sec}$$

$$\omega_m = 6.07 \text{ deg/sec}$$

Y/Minor Axis:

$$\frac{1}{2} \omega_s = 3.70 \text{ deg/sec}$$

$$\omega_s = 5.23 \text{ deg/sec}$$

$$\omega_m = 11.70 \text{ deg/sec}$$

The X-axis is the binding constraint, as it must arrest the inertia of the entire dish assembly in addition to its own counterweight system. All results carry explicit conservatism: component masses were raised by 20%, final inertia values were raised by a further 20%, and the torque values reflect rated continuous motor output rather than peak or instantaneous capability.

Even under the most conservative assumption (a single motor's rated torque on the slower X-axis), the calculated safe speed of 1.92°/sec is nearly six times the current operating limit of 0.33°/sec. Under the two-motor assumption, the X-axis limit of 2.71°/sec approaches the telescope's original hydraulic-era speed of 3°/sec, indicating that the new servo system is capable of performance comparable to the system it replaced.

The applicable result depends on the torque bias configuration of the dual-motor drive system on each axis. If both motors contribute net braking torque in the same direction during a stop, ω_s applies. If anti-backlash preloading reduces effective net torque to one motor's contribution, $\frac{1}{2}\omega_s$ is the appropriate limit. Clarification of the servopack torque bias parameters would allow a single definitive safe speed to be stated for each axis.

If a more precise moment of inertia becomes available through direct measurement, empirical testing, or a more detailed structural model, it can be substituted into $\omega_i = \sqrt{2\alpha\theta}$ without repeating any of the underlying work. The results presented here are a safe operational starting point, not a permanent ceiling.